NeuralMed

A Multi-Disease Clinical Screening Platform Using Classical ML and Deep Transfer Learning

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Overview

NeuralMed unifies **five disease-screening modules** and a **deep learning CT classifier** into a single Flask web application serving both patients and clinicians through dedicated portals.

Patient Portal — Algorithms

→ **Diabetes:** Gaussian Naïve Bayes — probabilistic classifier optimal for continuous clinical features (glucose, BMI, insulin)

→ **Heart Disease:** Naïve Bayes — strong performance on correlated cardiovascular markers (ECG, cholesterol, thalassemia)

→ **Parkinson's:** Random Forest (max_depth=10) — ensemble of decision trees on 8 vocal biomarkers; depth-constrained to prevent overfitting

→ **Breast Cancer:** Logistic Regression — linear separability in 30-feature FNA space; 98.2% test accuracy

→ **Kidney CT:** VGG-16 Transfer Learning — ImageNet pretrained CNN fine-tuned for 4-class CT classification (Normal/Cyst/Stone/Tumor)

Doctor Portal — Clinical Algorithm Outputs

→ **Confidence %:** model.predict_proba() — calibrated class probability

→ **Feature Importance:** RF feature_importances_ / LR coefficients — top-10 ranked by magnitude

→ **Reference Ranges:** dataset-derived benign vs. malignant thresholds displayed per input field

→ **Gemini 2.5 Flash Lite:** LLM structured prompt → Clinical Interpretation · Risk Factors · Next Steps · Caveats

Evaluation Protocol

→ 8 classifiers benchmarked per disease

→ Stratified 5-fold CV — mean ± std accuracy

→ Held-out 80/20 split — test acc, precision, recall, F1, AUC

System Architecture

DATA INPUT

Tabular: 5 clinical CSV datasets | Image: CT scan upload

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PREPROCESSING

StandardScaler · LabelEncoder · Median imputation Image: Resize 224×224 · Normalize [0,1] · Augmentation

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ALGORITHM LAYER

Disease	Algorithm	Why Chosen
Diabetes	GaussianNB	Best CV; handles small continuous features
Heart	Naive Bayes	Strong on correlated cardiac markers
Parkinson's	RF depth=10	Ensemble; max_depth=10 prevents overfitting
Breast Cancer	Logistic Reg.	Linear separability in FNA feature space
Kidney CT	VGG-16 TL	ImageNet weights; fine-tuned last layers

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OUTPUT LAYER

Patient Portal	Doctor Portal
Binary Prediction + Gemini Explanation	Confidence % · Feature Importance Reference Ranges · Clinical Summary

Algorithms & Evaluation

8 Classifiers Evaluated Per Disease

Naïve Bayes · Logistic Regression · k-NN · SVM (RBF) · Decision Tree · Random Forest · XGBoost · Extra Trees

5-Fold Stratified Cross-Validation

StratifiedKFold(n=5, shuffle=True, random_state=42) Reports: mean accuracy ± std, confusion matrix, ROC/AUC

Disease	Best Model	Test Acc	CV Mean	±Std	AUC
Breast Cancer	Logistic Reg.	**98.2%**	97.4%	1.7	**0.99**
Parkinson's	RF depth=10	**92.3%**	88.2%	4.8	**0.97**
Kidney CT	VGG-16 TL	**99.0%**	—	—	—
Heart Disease	Naive Bayes	85.2%	84.1%	4.3	0.92
Diabetes	GaussianNB	74.7%	76.0%	1.7	0.84

Overfitting Prevention

Random Forest and Decision Tree constrained to **max_depth=10** — prevents memorisation while maintaining competitive test accuracy and stable cross-validation scores.

CNN Architecture — Kidney CT

VGG-16 pretrained on ImageNet. Final dense layers replaced: GlobalAvgPool → Dense(256, ReLU) → Dropout(0.5) → Dense(4, Softmax). Classes: Normal · Cyst · Stone · Tumor. Dataset: 12,446 CT images, 80/10/10 split.

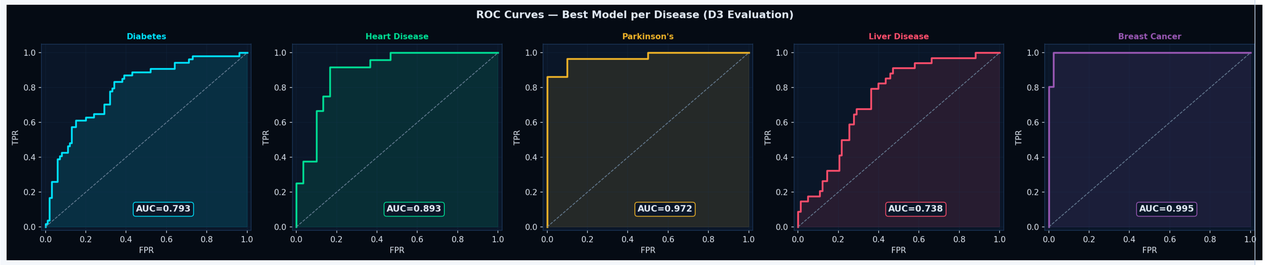


Fig. A. ROC Curves — AUC per disease

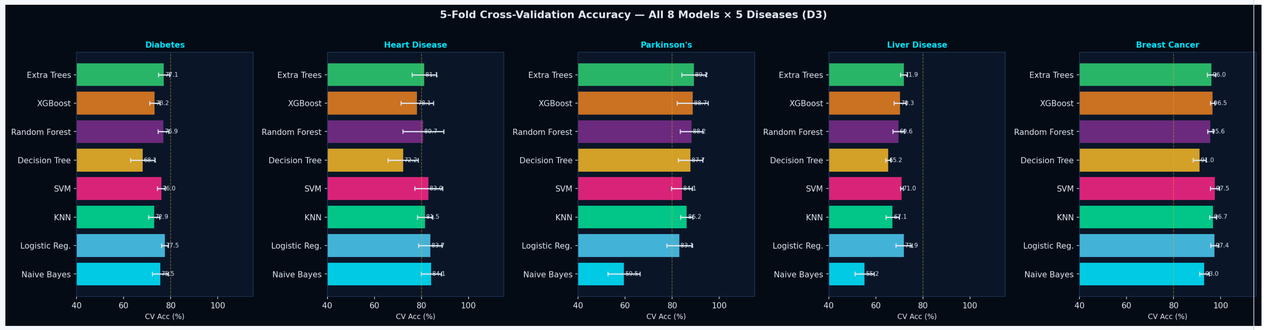
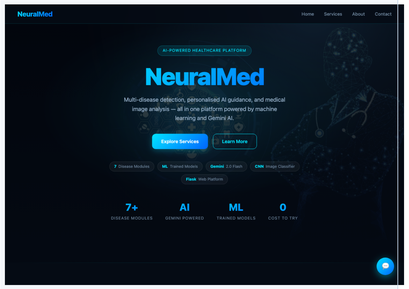
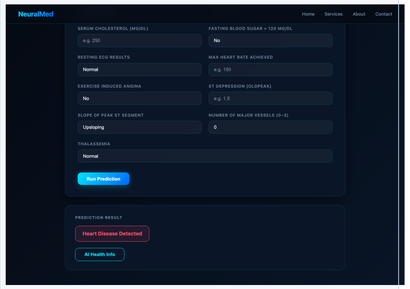


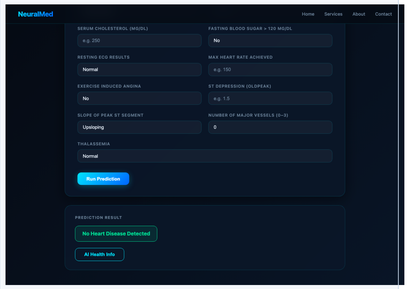
Fig. B. 5-Fold CV Accuracy — All Models × All Diseases



Landing — chatbot + modules



Heart Disease Detected — patient view



No Heart Disease Detected — patient view

Doctor Portal — Heart Disease Detected

Conclusion

What we built:

A production-ready Flask platform serving 5 disease-screening modules through dual clinical portals, with full evaluation pipeline.

Key results:

- ❖ Breast Cancer: 98.2% acc, AUC 0.99
- ❖ Kidney CT (VGG-16): 99.0% accuracy
- ❖ Parkinson's RF: 92.3% acc, 88.2% CV
- ❖ Heart Disease NB: 85.2% acc, AUC 0.92

Novel contributions:

- ❖ Unified multi-disease platform (5 modules, 2 portals)
- ❖ LLM-augmented clinical summaries via Gemini 2.5
- ❖ Depth-constrained ensemble with CV validation
- ❖ Doctor Portal with feature attribution + PDF export

Future work:

- ❖ Grad-CAM visualisations for CNN explainability
- ❖ SMOTE for class-imbalanced datasets
- ❖ Cloud deployment with ONNX runtime

References

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github.com/honeyreddydaram/NeuralMed

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